

Applications of Quiver Representations in Topological Data Analysis

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May 2025

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Abstract

Despite its many applications, finding a proper scale for topological data analysis proves to be a challenge (Obviously, we can use trial-and-error however, as logical and mortal beings, we wish to save time and automate the process). An algorithm for topological data analysis consists of creating filtration based on a parameter, discretizing them, applying the homology functor to form a module and studying its properties. This parameter may be height, time, resolution, etc. and this module is called *persistence module*. The study of these persistence modules is called *persistence theory*.

Quiver representations find their way in topological data analysis through the 'studying properties' path. As will be shown, every persistence module is equivalent to a *representation of a quiver with relations*. As such, all the decomposition theorems applying to quivers under certain conditions can also be applied to persistence modules. The aim of this project is to explore this particular aspect of persistence theory based on the book ***Persistence Theory: From Quiver Representations to Data Analysis*** by Steve Y. Oudot. We end the report with two calculations showing how two different approaches can be taken to study the action of homology functor on some filtration constructed over a dataset, the most general method being the one used in Example [3.2.1](#)

Introduction

One can easily count how many dots, holes, loops, clusters, etc. and of what kind (circular arrangement or linear arrangement of points) are there in data. However, how can a computer, which deals with numbers only, just "see" and know this information? This is made possible through data analysis. We study the algebraic topology approach of doing so, made popular by **Gunnar Carlsson** through his work "*Topology and Data*", 2009. He bridged the two different worlds of abstract topology and data analysis and introduced the idea of "shape" of data. Some other great contributors in this field are **Afra Zomorodian** and **Herbert Edelsbrunner**. They developed the theoretical framework of *persistent homology*, which basically entails the birth and death of features in a filtration. In this project, we aim to understand how this framework works for a simple dataset as given in Fig.3.1. The same method can be generalised to more complicated datasets.

The initial definition of *persistence modules* had been a finite dimensional representation of the linear quiver L_n (as per *Edelsbrunner, Letscher and Zomorodian*). As work progressed, the term was generalised to refer to any representation of a subset $T \subseteq \mathbb{R}$ which satisfy:

$$\begin{aligned} \forall i, v_i^i &= \text{id}_{V_i} \\ \forall i \leq j \leq k, v_i^k &= v_j^k \circ v_i^j \end{aligned}$$

, where $\{V_i\}_{i \in T}$ are vector spaces and $\{v_i^j : V_i \rightarrow V_j\}_{i \leq j \in T}$ are linear maps.

Through the one-sided equivalence of persistence modules with representations of quiver with relations, we get to carry over the decomposition theorems true for representation of quivers, obtained from works of **Krull, Remak, Schmidt, Auslander, Ringel, Tachikawa, Gabriel and Azumaya**, to persistence modules which gives us the theorem that persistence modules can be decomposed as a unique direct sum of interval modules (up to isomorphism and permutation), when either T is finite or T is arbitrary and the persistence module is pointwise finite dimensional.

Applying homology functor on a filtration indexed over $T \subseteq \mathbb{R}$ which is basically a nested sequence of topological spaces indexed over T , all of whose members are triangulable, we get pointwise finite dimensional persistence modules, which can be decomposed into a direct sum of indecomposable interval modules which give the birth and death of different features as connected components, holes, etc. We end the report with calculations that show two different methods of calculating the persistent homology (one algorithmic; one intuitive that cannot be performed by computer but is good to know).

Chapter 1

Basics of Quivers

1.1 What are Quivers?

A quiver in the English language is a portable case for holding arrows. Similarly, in mathematics, it is a "case" for holding arrows along with their source and target vertices, which are clearly defined through functions on the set of arrows. A precise description of quivers is provided in the following definition–

DEFINITION 1.1.1. A *quiver* $\mathcal{Q} = (Q_0, Q_1, s, t)$ is given by two set Q_0, Q_1 called the *vertex* set and *edge* set, respectively, and two mappings/functions $s, t : Q_1 \rightarrow Q_0$ called *source* and *target* functions, respectively. Graphically, $\mathcal{Q}$ is represented by a directed multigraph with self loops allowed with vertices indexed by the set Q_0 and arrows indexed by Q_1 directed from $s(\alpha) \rightarrow t(\alpha) \forall \alpha \in Q_1$.

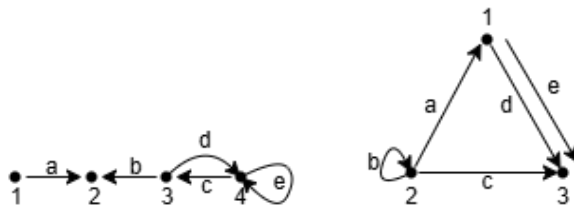


Figure 1.1: Examples of quivers

A quiver $\mathcal{Q}$ is called *finite* if Q_0 and Q_1 are finite (*viz.* number of arrows and vertices are finite). The quivers we will discuss from now on are the A_n -type quivers, whose underlying graph is finite and linear-shaped. The special case where all arrows are oriented to the right are called *linear quivers*, denoted L_n . From here on, we will be using the graph representation of quivers.

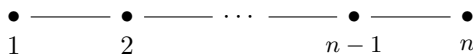


Figure 1.2: Underlying graph of an A_n -type quiver



Figure 1.3: An example of an A_5 - type quiver

1.2 Category of Quiver Representations

Consider two vector spaces and a linear transformation between them. It can be easily seen that a phenomenon of "going" from one vector space to another through the transformation is similar to a one-sided pathway from the first vector space to the second one, which is the same as an arrow (associated with the linear transformation) pointing from one node to another (nodes representing vector spaces). The following definitions and discussion will give more insight into this phenomenon—

DEFINITION 1.2.1. A *representation* of $\mathcal{Q}$ over a base field $\mathbf{k}$ is a pair $\mathbb{V} = (V_i, v_\alpha)$ consisting of a set of $\mathbf{k}$ -vector spaces $\{V_i | i \in Q_0\}$ together with a set of $\mathbf{k}$ -linear maps $\{v_\alpha : V_{s_\alpha} \rightarrow V_{t_\alpha} | \alpha \in Q_1\}$.

$$\begin{array}{ccccccc} \begin{pmatrix} 1 \\ 0 \end{pmatrix} & & \begin{pmatrix} 1 \\ 1 \end{pmatrix} & & \begin{pmatrix} 0 \\ 1 \end{pmatrix} & & \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ k \longrightarrow k^2 \longleftarrow k \longrightarrow k^2 \longrightarrow k^2 & & k \xrightarrow{0} 0 \xleftarrow{0} k \xrightarrow{1} k \xrightarrow{0} 0 \end{array}$$

Figure 1.4: Some representations of quiver shown in Figure 1.3

A representation $\mathbb{V}$ is *finite dimensional* if the sum of dimensions of its constituent spaces V_i is finite. Henceforth, we will be fixing the base field as $\mathbf{k}$.

DEFINITION 1.2.2. A representation $\mathbb{W} = (W_i, w_\alpha)$ is a *subrepresentation* of $\mathbb{V} = (V_i, v_\alpha)$ if W_i is a subspace of V_i for all $i \in Q_0$ and if w_α is the restriction of the map v_α to the subspace W_{t_α} for all $\alpha \in Q_1$.

DEFINITION 1.2.3. A *morphism* ϕ between two $\mathbf{k}$ -representations $\mathbb{V}, \mathbb{W}$ of $\mathcal{Q}$ is a set of $\mathbf{k}$ -linear maps $\phi_i : V_i \rightarrow W_i$ such that the following diagram commutes for every arrow $\alpha \in Q_1$:

$$\begin{array}{ccc} V_{s_\alpha} & \xrightarrow{v_\alpha} & V_{t_\alpha} \\ \phi_{s_\alpha} \downarrow & & \downarrow \phi_{t_\alpha} \\ W_{s_\alpha} & \xrightarrow{w_\alpha} & W_{t_\alpha} \end{array}$$

$$\begin{array}{ccccccc} \begin{pmatrix} 1 \\ 0 \end{pmatrix} & & \begin{pmatrix} 1 \\ 1 \end{pmatrix} & & \begin{pmatrix} 0 \\ 1 \end{pmatrix} & & \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ k \longrightarrow k^2 \longleftarrow k \longrightarrow k^2 \longrightarrow k^2 & & k \xrightarrow{0} 0 \xleftarrow{0} k \xrightarrow{1} k \xrightarrow{0} 0 \\ \downarrow 1 & & \downarrow 0 & & \downarrow 1 & & \downarrow \begin{pmatrix} 0 & 1 \end{pmatrix} \\ k \longrightarrow 0 \longleftarrow k \longrightarrow k \longrightarrow 0 & & & & & & \downarrow 0 \end{array}$$

Figure 1.5: An example of morphism between representations shown in Figure 1.4

The morphisms between representations can be composed pointwise (i.e. at each vertex independently). Namely, if $\phi : \mathbb{U} \rightarrow \mathbb{W}$ and $\psi : \mathbb{W} \rightarrow \mathbb{V}$ then $(\psi \circ \phi)_i = \psi_i \circ \phi_i \forall i \in Q_0$. And we also have

an identity morphism $\mathbb{1}_{\mathbb{V}} : \mathbb{V} \rightarrow \mathbb{V}$ defined as $(\mathbb{1}_{\mathbb{V}})_i = \mathbb{1}_{V_i} \forall i \in Q_0$. This turns the representation of $\mathcal{Q}$ into a category, called $\mathcal{Rep}_k \mathcal{Q}$. The subcategory of finite dimensional quivers is called $rep_k \mathcal{Q}$. Both these categories are abelian (*abelian category* is a category in which morphisms and objects can be added and in which kernels and cokernels exist and have desirable properties), in particular:

- The zero object is the *trivial representation* with all spaces and all maps equal to 0.

$$0 \xrightarrow{0} 0 \xleftarrow{0} 0 \xrightarrow{0} 0 \xrightarrow{0} 0$$

Figure 1.6: Trivial representation of quiver given in Figure 1.3

- There is a biproduct, the *direct sum* defined for any $\mathbb{V}, \mathbb{W}$ as $\mathbb{V} \oplus \mathbb{W}$ with spaces $V_i \oplus W_i \forall i \in Q_0$ and maps $v_\alpha \oplus w_\alpha = \begin{pmatrix} v_\alpha & 0 \\ 0 & w_\alpha \end{pmatrix} \forall \alpha \in Q_1$. If a nontrivial representation is isomorphic to direct sum of nontrivial representations, called *summands*, then it is called *decomposable*, otherwise, *indecomposable*.

$$k^2 \xrightarrow{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}} k^2 \xleftarrow{\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}} k^2 \xrightarrow{\begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}} k^3 \xrightarrow{\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}} k^2$$

Figure 1.7: Direct sum of representations in Figure 1.4

- Every morphism $\phi : \mathbb{V} \rightarrow \mathbb{W}$ has a *kernel*, *cokernel* and *image* defined pointwise (as in, $(\ker \phi)_i = \ker \phi_i \forall i \in Q_0$ and similarly, for cokernel and image).¹

$$0 \xrightarrow{0} k^2 \xleftarrow{0} 0 \xrightarrow{0} k \xrightarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} k^2$$

Figure 1.8: Kernel of morphism in Figure 1.5

- A morphism $\phi : \mathbb{V} \rightarrow \mathbb{W}$ is injective (*resp.* surjective) if it is pointwise injective (*resp.* surjective) that is ϕ_i are injective (*resp.* surjective) maps from V_i to W_i for all $i \in Q_0$.

Despite these similarities with vector spaces, quiver representations is also different in some aspects, for example, *semisimplicity*. (All subspaces of a vector space have a complement subspace but this is not true in general for representation. For example, if $\mathbb{V} = k \xrightarrow{1} k$ and $\mathbb{W} = 0 \xrightarrow{0} k$, then $\mathbb{W}$ is a subrepresentation of $\mathbb{V}$ but there is no subrepresentation $\mathbb{U}$ of $\mathbb{V}$ such that $\mathbb{V} = \mathbb{U} \oplus \mathbb{W}$.)

1.3 Classification of Quiver Representations

Now, as with every mathematical object, we also aim to classify quiver representations. Not all can be classified properly, however, those of the Dynkin type, especially *A_n-type* quivers, can be and in this section, we will be discussing theorems related to this.

¹The kernel, cokernel and image are isomorphic to subrepresentations of $\mathbb{V}, \mathbb{W}$ and $\mathbb{W}$, respectively.

THEOREM 1.3.1. *A representation $\mathbb{V}$ of $\mathcal{Q}$ is indecomposable if and only if for all idempotent morphisms $\phi : \mathbb{V} \rightarrow \mathbb{V}$ ², we have $\phi = 0$ or $\phi = \mathbb{1}_{\mathbb{V}}$.*

THEOREM 1.3.2. [2] (Krull, Remak, Schmidt) *Let $\mathcal{Q}$ be a finite quiver, and let $\mathbf{k}$ be a field. Then, every finite-dimensional representation $\mathbb{V}$ of $\mathcal{Q}$ over $\mathbf{k}$ decomposes as a direct sum:*

$$\mathbb{V} = \mathbb{V}^1 \oplus \cdots \oplus \mathbb{V}^r,$$

where each $\mathbb{V}^i$ is itself indecomposable i.e. cannot be further decomposed into a direct sum of at least two nonzero representations. Moreover, the decomposition is unique up to isomorphism and permutation of the terms in the direct sum.

THEOREM 1.3.3. [2] (Gabriel for A_n -type quivers) *Let $\mathcal{Q}$ be an A_n -type quiver, and let $\mathbf{k}$ be a field. Then, every indecomposable finite-dimensional representation of $\mathcal{Q}$ over $\mathbf{k}$ is isomorphic to some interval representation $\mathbb{I}_{\mathcal{Q}}[b, d]$, described as follows:*

$$\begin{array}{cccccccccccccccc} 0 & \xrightarrow{0} & \cdots & \xrightarrow{0} & 0 & \xrightarrow{0} & \mathbf{k} & \xrightarrow{1} & \cdots & \xrightarrow{1} & \mathbf{k} & \xrightarrow{0} & 0 & \xrightarrow{0} & \cdots & \xrightarrow{0} & 0 \\ 1 & & & & b-1 & & b & & & & d & & d+1 & & & & n \end{array}$$

Therefore, every finite-dimensional representation of an A_n -type quiver $\mathcal{Q}$ decomposes uniquely (up to isomorphism and permutation of the terms) as a direct sum of interval representations.

We wish to classify infinite-dimensional representations of *infinite extensions of linear quivers*, with underlying graph as a set of nodes (countable) indexed by some poset, the reason for which will be mentioned at the end of this section. However, this is not possible in general. So, we restrict ourselves to minimising the number of constraints on the representation to be classified. We first give a theorem to classify infinite-dimensional representations of A_n -type quivers, the only constraint here being finiteness of the quiver (not dimension of the vector spaces). This will be followed by a theorem aimed at classifying *linear* quivers indexed by $\mathbb{N}$ or $\mathbb{Z}$, with the constraints here being the indexing over integers and pointwise finite-dimensionality (every vector space must be finite dimensional) and the *linearity* of quiver must hold.

THEOREM 1.3.4. [2] (Auslander, Ringel, Tachikawa, Gabriel, Azumaya) *Let $\mathcal{Q}$ be an A_n -type quiver and let $\mathbf{k}$ be a field. Then, every indecomposable representation of $\mathcal{Q}$ over $\mathbf{k}$ is isomorphic to an interval representation $\mathbb{I}_{\mathcal{Q}}[b, d]$ and every representation of $\mathcal{Q}$, whether finite- or infinite-dimensional, is isomorphic to a (possibly infinite) direct sum of interval representations. Moreover, this decomposition is unique up to isomorphism and permutation of the terms.*

Before stating the next theorem, we will first mention what we mean by infinite extensions of linear quivers indexed by $\mathbb{N}$ or $\mathbb{Z}$ (only for the sake of complete clarity; otherwise, it is self-explanatory).

² ϕ being idempotent means $\phi^2 = \phi$

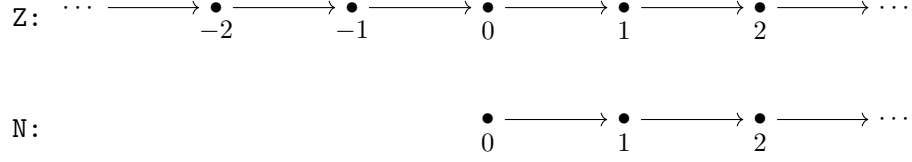


Figure 1.9: Infinite extensions of L_n – type quivers indexed by $\mathbb{Z}$ and $\mathbb{N}$ respectively. The representations of $\mathbb{N}$ quiver can be thought of as a representation of $\mathbb{Z}$ with vector spaces V_i indexed by $i < 0$ to be 0 and the maps between them and from V_{-1} to V_0 to be 0

THEOREM 1.3.5. [2](Webb) *Let $\mathbf{k}$ be a field. Then, any pointwise finite-dimensional representation of $\mathbb{Z}$ over $\mathbf{k}$ is a direct sum of interval representations.*

The main aim of our project is to explore the application of quiver representations in topological analysis. However, in topological analysis, we deal with filtration over sublevel or superlevel sets, which are generally indexed over a subset of real numbers and not finite. In order to decompose representations of such indexed sets (or, in general, posets), quiver representations, along with other theorems and tools, provide a very powerful tool. Before explaining the reason for this, some more knowledge is needed.

Representations of arbitrary subposets of $(\mathbb{R}, \leq)$. Take an arbitrary subset T of $\mathbb{R}$ and define a poset $(T, \leq)$. This poset forms a category with every element $i \in T$ as an object and a single morphism per couple $i \leq j$. We define a *representation of $(T, \leq)$* to be a functor to the category of vector spaces i.e. it defines vector spaces $(V_i)_{i \in T}$ and linear maps $(v_i^j : V_i \rightarrow V_j)_{i \leq j \in T}$. By functoriality, these maps satisfy:

$$\begin{aligned} v_i^i &= \mathbb{1}_{V_i} & \forall i \in T, \text{ and} \\ v_i^k &= v_j^k \circ v_i^j & \forall i \leq j \leq k \in T \end{aligned} \tag{1.1}$$

This definition can be used to defined a representation of any given poset.

Now, moving towards our goal, we state a theorem by Crawley-Boevey for representations of arbitrary subposets of $(\mathbb{R}, \leq)$ with pointwise finite dimensionality being the only constraint.

THEOREM 1.3.6. [2] (Crawley-Boevey) *Let $\mathbf{k}$ be a field and let $T \subseteq \mathbb{R}$. Then, any pointwise finite-dimensional representation of $(T, \leq)$ over $\mathbf{k}$ is a direct sum of interval representations.*

Now, we remark on the connection between category of quivers (with one object per node and one morphism per oriented path) and of posets. It wouldn't be accurate to say that these two are equivalent as such since paths sharing the same source and target are considered same in posets due to transitivity, but not in quivers. However, by defining relations on quivers that make such paths equal, we can make any poset equivalent (as a category) to some *quiver with relations*, and its representations can, thus, be viewed as quiver representations.

1.4 A Step Towards Persistence Theory

To not give this whole discussion a placebo effect, we wish to note the relevance of all these results to *persistence modules* but we must first define what interval modules and persistence modules are

and a few of their properties that must be assumed to make them 'well-behaved', as we will see (not without relevance, of course). The details of persistence theory shall be discussed in the next section.

DEFINITION 1.4.1. A *zigzag module* of length n is a representation of an A_n -type quiver. We can also define it as a representation of a poset $(\{1, 2, \dots, n\}, \preceq)$ whose *Hasse diagram* is of A_n -type.

³

DEFINITION 1.4.2. Given a poset $(T, \leq)$ where $T \subseteq \mathbb{R}$ and $\leq$ is the normal ordering on $\mathbb{R}$, a *persistence module* is a representation of the poset. This is different from zigzag modules in the sense that there is no restriction on the index being a finite set of integers up to n and $\preceq$ is by definition, $\leq$, unlike in zigzag modules.

An *interval module* is, in effect, similar to an interval decomposition of an A_n -type quiver. From this point on, we will take T as a subset of $\mathbb{R}$.

DEFINITION 1.4.3. An *interval* S of an arbitrary index set $T \subseteq \mathbb{R}$ is a subset $S \subseteq T$ such that if $i \leq j$ where $i, j \in S$ and $\exists k$ such that $i \leq k \leq j$, then $k \in S$. Note here, $\leq$ is the obvious ordering on $\mathbb{R}$, oblivious to the partial ordering defined by $\preceq$ (which will be the same as $\leq$ for persistence modules).

DEFINITION 1.4.4. Given an interval S of T , the associated *interval module* is a representation of the poset $(T, \preceq)$ in which the vector space V_i is $\mathbf{k}$ if $i \in S$, else $V_i = 0$ and the linear map $v_i^j = id$ if $i, j \in S$, else $v_i^j = 0$.

We have already discussed that the representation of any subposet of $(\mathbb{R}, \leq)$ is equivalent to a *quiver with relations*. Hence, the theorems for interval decomposition discussed earlier for quivers will naturally apply to zigzag modules and persistence modules, subject to certain conditions. This is specified in the following theorem.

THEOREM 1.4.1. (Interval Decomposition) *Given a poset $(T, \preceq)$ where $\preceq$ is an arbitrary partial order relation on T , we can decompose a representation $\mathbb{V}$ of it into a direct sum of interval modules, each of which is indecomposable, and this direct sum is unique up to permutation and isomorphism, whenever:*

1. T is finite and $\mathbb{V}$ is a zigzag module.
2. T is arbitrary, $\preceq$ is $\leq$, $\mathbb{V}$ is a pointwise finite-dimensional persistence module.

Now that we have obtained theorems stating decomposability of representations of arbitrary posets $(T, \preceq)$ where $T \subseteq \mathbb{R}$, we move on to the crux of the matter, how does one obtain a persistence module from given topological data.

³A Hasse diagram is a graph having vertices $\{1, 2, \dots, n\}$ and a single directed edge $i \rightarrow j$ whenever $i \prec j$ and $\nexists k \mid i \prec k \prec j$.

Chapter 2

Introduction to Homological Algebra

After having discussed persistence modules and theorems relating to their decomposition, we wish to study persistence diagrams of the evolution of some topological structure against some parameter (The exact meaning will be defined in the next chapter). For this purpose, an understanding of basic *simplicial homology*, a description of *homology functors* and *triangulation* is necessary. We will first start with giving a mathematical description of abstract simplicial complexes and then, we will move on to a geometric realization. The aim of this chapter is to make the reader familiar with calculations that bring us from a topological structure to a persistence module so that the evolution of the original structure with relation to the parameter can be predicted.

2.1 Simplicial Complexes- An Abstract Perspective

A simplicial complex is a finite set of subsets of a given set X that contains the power set of all its elements. A more precise definition is provided:

DEFINITION 2.1.1. Given a set X , an *abstract simplicial complex* is a family Δ of finite subsets of X such that if $\tau \in \Delta$ and $\sigma \subseteq \tau$, then $\sigma \in \Delta$. The elements of Δ are called *faces* of Δ , and the dimension of a face τ is $|\tau| - 1$. If $\sigma \subseteq \tau$, then we say that σ is a face of τ . A face τ is a *maximal face* of Δ if there is no face σ of Δ such that $\tau \subsetneq \sigma$. The 0-dimensional simplices are called *vertices* and 1-dimensional simplices are called *edges*. A simplicial complex Δ_0 is a simple and loopless *graph* if it has only vertices and edges. A simplicial complex Δ_0 is a *subcomplex* of Δ if $\Delta_0 \subseteq \Delta$. For $k \geq -1$, the *k-skeleton* $\Delta^{(k)}$ of Δ is the subcomplex of Δ obtained by removing all faces of dimension greater than k .

Seeing the terms *vertices* and *edges*, the reader may wonder if a simplicial complex has a geometric equivalent, to which the answer is yes. Rather than an equivalent, it is a geometric realization of simplicial complexes, which will be more precisely stated henceforth.

2.2 Geometric Realizations of Simplicial Complexes

A simplicial complex can be realised as a geometric object in $\mathbb{R}^n$. Vertices can be taken as points and edges as line segments between the respective labelled points (labelled by the index of vertices). A mathematical description for geometric realisation of a general n -dimensional simplex is given: For any $d \geq 0$, the *standard d -simplex* to be the set

$$X_d = \{(\lambda_0, \dots, \lambda_d) : \lambda_i \geq 0 \text{ for } 0 \leq i \leq d, \lambda_0 + \dots + \lambda_d = 1\} \subset \mathbb{R}^{d+1}.$$

In other words, X_d is the *convex hull* of the *standard basis* of $\mathbb{R}^{d+1}$.¹ One can define a mapping $f : V \rightarrow \mathbb{R}^n$ on the vertex set V of Δ . This f induces a map $f_\sigma : X_d \rightarrow \mathbb{R}^n$ by

$$f_\sigma(\lambda_0, \dots, \lambda_d) = \lambda_0 f(a_0) + \dots + \lambda_d f(a_d)$$

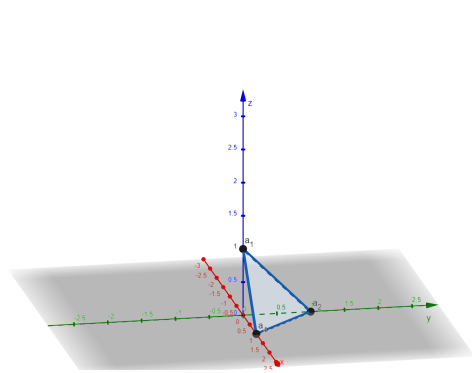
f induces a *geometric realization* of Δ if the following hold:

- The map f_σ is injective for each $\sigma \in \Delta \setminus \{\emptyset\}$. This property is meant to ensure that a higher dimensional object does not collapse to a lower dimensional object under the transformation (such as a 2-simplex realization, triangle, to a 1-simplex realization, line).
- For any nonempty $\sigma, \tau \in \Delta$,

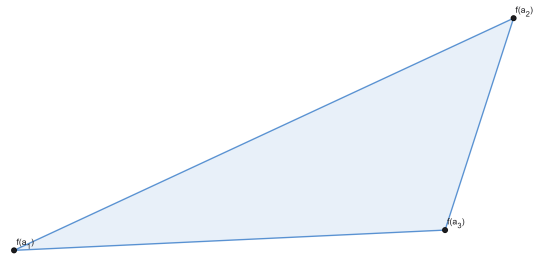
$$\text{im } f_\sigma \cap \text{im } f_\tau = \text{im } f_{\sigma \cap \tau},$$

where $\text{im } f$ denotes the image under f . Also, we define $f_\emptyset = \emptyset$. This property implies geometrically that the intersection of the realizations of any two faces contains nothing more than the realization of the greatest common face of the two simplices. It also implies that the intersection of geometric form of two simplices also represents a simplex.

The actual geometric realization is $|\Delta| = \bigcup_{\sigma \in \Delta \setminus \{\emptyset\}} \text{im } f_\sigma$



(a) A geometric realization of a 2-simplex



(b) Transformation of Fig 2.1a by f_σ mapping $(1, 0, 0)$ to $f(a_1)$, $(0, 1, 0)$ to $f(a_2)$ and $(0, 0, 1)$ to $f(a_3)$

Source: geogebra.org

Figure 2.1

¹The convex hull of a set of points is the smallest convex set containing those points. If S is a set of points in $\mathbb{R}^n$, then $\text{conv}(S) = \{\sum \lambda_i p_i | \lambda_i \geq 0 \forall i, \text{ and } \sum \lambda_i = 1\}$

From here on out, we will map the vertices $v_1, \dots, v_k$ to standard basis $\mathbf{e}_i$ of $\mathbb{R}^k$.

2.3 Boundary Maps

First, we must define what *chain groups* are.

DEFINITION 2.3.1. Let $\mathbb{F}$ be a commutative ring and Δ be a simplicial complex. A *reduced chain group* $\tilde{C}_n(\Delta; \mathbb{F})$ is a free $\mathbb{F}$ -module whose basis are indexed by the n -dimensional faces of Δ . In other words, for each face $a_0 a_1 \dots a_n$, there is a basis element $\mathbf{e}_{a_0 a_1 \dots a_n}$. A basis element is an *oriented simplex*. An *unreduced chain group* $C_n(\Delta; \mathbb{F})$ is one which agrees with $\tilde{C}_n(\Delta; \mathbb{F})$ everywhere, except at $n = -1$ where it is 0.

Wherever the ring $\mathbb{F}$ is obvious, we will write $\tilde{C}_n(\Delta; \mathbb{F})$ as $\tilde{C}_n(\Delta)$. To make sure that only one basis element is defined for a face, we let $\mathbf{e}_{ab}$ (where a and b are oriented 0-simplices) satisfy-

$$\begin{aligned}\mathbf{e}_{ba} &= -\mathbf{e}_{ab} \\ \mathbf{e}_{aa} &= 0\end{aligned}$$

The higher dimensional faces also define unique basis elements of their respective chain groups due to the first rule. For example:

$$\mathbf{e}_{dbca} = -\mathbf{e}_{dbac} = \mathbf{e}_{dabc} = -\mathbf{e}_{adbc} = \mathbf{e}_{abdc} = -\mathbf{e}_{abcd}$$

The second rule is mostly for geometric significance.

Now, we are ready to define *boundary maps*.

DEFINITION 2.3.2. A *boundary map* d_n , is a function $d_n : \tilde{C}_n(\Delta) \rightarrow \tilde{C}_{n-1}(\Delta)$ is a linear map $d_n(\mathbf{e}_{a_0 a_1 \dots a_n}) = \sum_{r=0}^n (-1)^r \mathbf{e}_{a_0 \dots a_{r-1} \hat{a}_r a_{r+1} \dots a_n}$ for each case n , where $\hat{a}_r$ denotes removal of element a_r . As a special case, $d_0(\mathbf{e}_a) = \mathbf{e}_\emptyset$ for each vertex a .

Note here that boundary maps satisfy $d_n \circ d_{n+1} = 0$ for every n , which is what we would expect geometrically. The derivation for this is given in [1]. We are leaving it out here and stating this property as a fact. This implies that the sequence

$$\mathcal{C}(\Delta) : \quad \dots \xrightarrow{d_{n+2}} \tilde{C}_{n+1}(\Delta) \xrightarrow{d_{n+1}} \tilde{C}_n(\Delta) \xrightarrow{d_n} \tilde{C}_{n-1}(\Delta) \xrightarrow{d_{n-1}} \dots$$

defines a *chain complex*. The complex $\mathcal{C}(\Delta)$ is called *simplicial chain complex*. Also, a geometric interpretation of a boundary map of a given edge ab is a line segment oriented from a to b , we define

$$d_1(\mathbf{e}_{ab}) = \mathbf{e}_b - \mathbf{e}_a$$

and so on for other edges. We may also see $\mathbf{e}_{ab}$ as a vector $\mathbf{v}$ oriented from $\mathbf{e}_a$ to $\mathbf{e}_b$. This perspective will prove to be useful when calculating rank of kernel and image of boundary maps in certain cases as will be shown in Example 2.5.2

2.4 Homology Groups

Now that we have discussed simplicial chain complexes, we come to the topic of how one can gain useful information out of them. Information like number of *connected components*, *holes* and *voids* and even higher dimensional holes which albeit cannot be visualised (for we live in a 3d space) but still useful in some areas of study can be obtained through the algebraic formula of *homology groups*. Before doing so, we first direct the reader towards a new and perhaps, unfamiliar perspective of holes. The reader may see holes and voids as closed structures (in other words, having boundary or image under boundary map 0), called *cycles*, which have empty interiors or in other words, are themselves not a *boundary* of some structure contained in the simplicial complex. Now that we have suggested this, we write this in mathematical terms. For this, we must first define the $\mathbb{F}$ -module $Z_n(\Delta; \mathbb{F})$ of cycles and the $\mathbb{F}$ -module $B_n(\Delta; \mathbb{F})$ of boundaries by the formulas:

$$\begin{aligned} Z_n(\Delta; \mathbb{F}) &:= \ker d_n \\ &= \{z \in \tilde{C}_n(\Delta; \mathbb{F}) : d_n(z) = 0\}, \\ B_n(\Delta; \mathbb{F}) &:= \text{im } d_{n+1} \\ &= \{z \in \tilde{C}_n(\Delta; \mathbb{F}) : z = d_{n+1}(x) \text{ for some } x \in \tilde{C}_{n+1}(\Delta; \mathbb{F})\}. \end{aligned}$$

DEFINITION 2.4.1. The *simplicial homology in degree n of Δ* is given by:

$$\tilde{H}_n(\Delta; \mathbb{F}) = Z_n(\Delta; \mathbb{F}) / B_n(\Delta; \mathbb{F}).$$

The members of $\tilde{H}_n(\Delta; \mathbb{F})$ are called *homology classes*; each such member is an equivalence class of the relation $z \sim z' \iff z - z' \in B_n(\Delta; \mathbb{F})$; The *tilde* symbol here indicates *reduced homology* defined over reduced chain groups. *Unreduced homology* $H_n(\Delta; \mathbb{F})$ is also defined using the same formula on unreduced chain groups.

Note that quotienting two rings is basically "killing the denominator". The rank of $H_0(\Delta; \mathbb{F})$ gives the number of *connected components*, the derivation for which may be found in [1]. In 3d spaces, we mostly have to calculate $H_0(\Delta; \mathbb{F})$, $H_1(\Delta; \mathbb{F})$ and $H_2(\Delta; \mathbb{F})$ as they represent number of connected components, holes and voids respectively. The homology groups for $n > 2$ has rank 0. The proof is trivial. We leave this as an exercise for the reader.

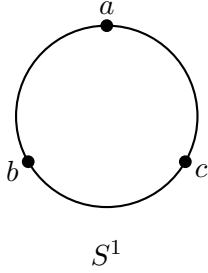
2.5 Triangulation of Topological Spaces

This is where the real geometry happens. We will only deal with triangulation by simplicial complexes.

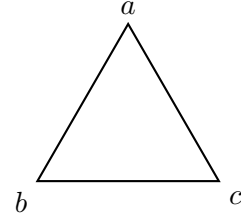
DEFINITION 2.5.1. A *triangulation* of a topological space $\mathbb{X}$ is a simplicial complex K such that there is a homeomorphism between $|K|$ and X .

We will proceed by giving a few examples. These examples are not exhaustive by any means, however they are enough to give the reader an idea of how a triangulation of a topological space can be formed and how the homology functors are applied to them. From now on, we consider $\mathbb{F}$ to be a *field*, so that the rank-nullity theorem holds for boundary maps.

Example 2.5.1. We take the topological space $S^1 = \{(x, y) | x^2 + y^2 = 1\}$. We obtain a triangulation as $K = \{\emptyset, a, b, c, ab, bc, ac\}$. We can see that S^1 and $|K|$ are homeomorphic as the vertices of $|K|$ are mapped to points on the circumference of the circle and the *edges* are mapped to respective arcs of S^1 as shown in the figure.



(a) The circle S^1 with labeled points



(b) Geometric realization of a triangulation of S^1

Figure 2.2

The chain groups for K are $C_{-1}(K; \mathbb{F}) = 0$, $C_0(K; \mathbb{F}) = \mathbb{F}^3$, $C_1(K; \mathbb{F}) = \mathbb{F}^3$ and $C_n(K; \mathbb{F}) = 0 \forall n \geq 2$. The simplicial chain complex is:

$$\cdots \xrightarrow{0} 0 \xrightarrow{0} \mathbb{F}^3 \xrightarrow{d_1} \mathbb{F}^3 \xrightarrow{d_0} 0 \xrightarrow{0} \cdots$$

From the chain complex, we can already infer that $\ker(d_0) = \mathbb{F}^3$, $\text{im}(d_0) = 0$, $\text{im}(d_n) = 0 \forall n \geq 2$ and $\ker(d_n) = 0 \forall n \geq 2$. Now, we only have calculate the rank of $\text{im}(d_1)$ and $\ker(d_1)$. A general element in $C_1(K; \mathbb{F})$ is of the form $x = \lambda_1 \mathbf{e}_{ab} + \lambda_2 \mathbf{e}_{bc} + \lambda_3 \mathbf{e}_{ac}$ where $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{F}$. Applying d_1 to it, we get:

$$d_1(x) = (\lambda_1 + \lambda_3) \mathbf{e}_a + (\lambda_2 - \lambda_1) \mathbf{e}_b + (-\lambda_2 - \lambda_3) \mathbf{e}_c$$

. This can only be 0 when $\lambda_1 = \lambda_2 = -\lambda_3$. Hence $\ker(d_1) = \langle \mathbf{e}_{ab} + \mathbf{e}_{bc} - \mathbf{e}_{ac} \rangle$ and hence, the rank of kernel is 1. By *rank-nullity theorem* for free $\mathbb{F}$ -modules, rank of image of d_1 is 2.

Now, we calculate the homology groups.

$$H_0(K; \mathbb{F}) = \frac{\ker(d_0)}{\text{im}(d_1)} \cong \frac{\mathbb{F}^3}{\mathbb{F}^2} \cong \mathbb{F}$$

$$H_1(K; \mathbb{F}) = \frac{\ker(d_1)}{\text{im}(d_2)} \cong \frac{\mathbb{F}}{0} \cong \mathbb{F}$$

$$H_n(K; \mathbb{F}) = 0 \forall n \geq 2$$

Notice, how the rank of $H_0(K; \mathbb{F})$ is the same as the number of connected components. This remains true in even higher dimensions, as stated earlier. Moreover, the rank of $H_1(K; \mathbb{F})$ is equal to the number of holes. This is not true in general, however, it is true for triangulations of topological spaces that can be embedded in $\mathbb{R}^3$.

Example 2.5.2. Take the topology space $S^1 \times I^1$ which is a cylinder geometrically. A triangulation of this space is given by $\triangle = \{\emptyset, v_0, v_1, v_2, v_3, v_4, v_5, v_0v_1, v_1v_2, v_2v_3, v_3v_4, v_4v_5, v_5v_0, v_0v_3, v_1v_4, v_2v_5,$

$v_1v_3, v_2v_4, v_0v_5, v_0v_1v_3, v_1v_3v_4, v_1v_2v_4, v_2v_4v_5, v_0v_2v_5, v_0v_3v_5\}$ whose geometric realization $|\Delta|$ is given in Fig. 2.3b. The vertices v_0, v_1 and v_2 are mapped to three distinct points on the upper circular rim of the cylinder while v_3, v_4 and v_5 are mapped to three distinct points on the lower circular rim of the cylinder under the action of homeomorphism.

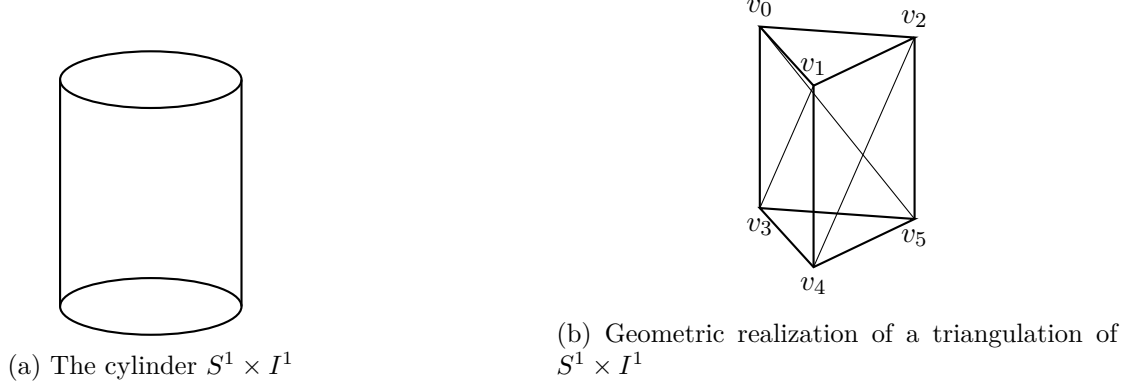


Figure 2.3

The simplicial chain complex is given by:

$$\cdots \xrightarrow{0} 0 \xrightarrow{0} \mathbb{F}^6 \xrightarrow{d_2} \mathbb{F}^{12} \xrightarrow{d_1} \mathbb{F}^6 \xrightarrow{d_0} 0 \xrightarrow{0} \cdots$$

From the chain complex, we can conclude $\ker(d_0) = \mathbb{F}^6$, $\text{im}(d_0) = 0$, $\ker(d_n) = 0 \forall n \geq 3$ and $\text{im}(d_n) = 0 \forall n \geq 3$. For calculating rank of kernel and image of d_1 and d_2 , we use the vector perspective we had mentioned earlier. This will give us something as shown in Fig 2.4

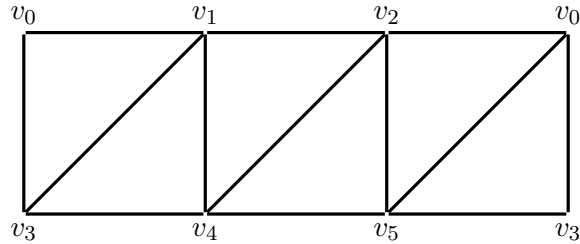


Figure 2.4: The geometric realization of the simplicial complex Δ laid out

In order to find rank of $\text{im}(d_1)$, we write the result of $d_1(x)$ where x is a basis element of $C_1(\Delta; \mathbb{F})$ in matrix form as in Fig. 2.5a. Here, the labels of the columns and rows are indices of the basis elements of $C_1(\Delta; \mathbb{F})$ and $C_0(\Delta; \mathbb{F})$. The column entries give the result of d_1 on the basis elements. For example, $d_1(\mathbf{e}_{v_0v_1}) = \mathbf{e}_{v_1} - \mathbf{e}_{v_0}$, so the column entries of the column labelled v_0v_1 are -1 and 1 for rows v_0 and v_1 . Now, we perform column operations on the matrix obtained and reduce it to the matrix given in Fig. 2.5b. Hence, $\text{im}(d_1) = -\lambda_1v_0 + (\lambda_1 - \lambda_2)v_1 + (\lambda_2 - \lambda_5)v_2 + (-\lambda_3 + \lambda_5)v_3 + (\lambda_3 - \lambda_4)v_4 + \lambda_4v_5$ where $\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5 \in \mathbb{F}$. Hence, rank of $\text{im}(d_1)$ and $\ker(d_1)$ are 5 and 7 respectively (by rank-nullity theorem).

	v_0v_1	v_1v_2	v_0v_2	v_3v_4	v_4v_5	v_3v_5	v_0v_3	v_1v_4	v_2v_5	v_1v_3	v_2v_4	v_0v_5
v_0	-1		-1				-1					-1
v_1	1	-1						-1		-1		
v_2		1	1						-1		-1	
v_3				-1		-1	1			1		
v_4				1	-1			1			1	
v_5					1	1			1			1

(a)

	v_0v_1	v_1v_2	v_3v_4	v_4v_5	v_0v_3
v_0	-1				
v_1	1	-1			
v_2		1			-1
v_3			-1		1
v_4			1	-1	
v_5				1	

(b)

Figure 2.5

We call this boundary matrix of d_2 . Similarly, we calculate rank of image and kernel of d_2 by constructing matrix in a similar fashion. We obtain the matrix given in Fig.2.6. One can already see that the column vectors so obtained are linearly independent. Hence, the rank of $im(d_2)$ is the number of column vectors, which is 6 and by rank-nullity theorem, the rank of $ker(d_2)$ is $12 - 6 = 6$.

	$v_0v_1v_3$	$v_1v_3v_4$	$v_1v_2v_4$	$v_2v_4v_5$	$v_0v_2v_5$	$v_0v_3v_5$
v_0v_1	1					
v_1v_2			1			
v_0v_2					1	
v_3v_4		1				
v_4v_5				1		
v_3v_5						1
v_0v_3	-1					1
v_1v_4		-1	-1			
v_2v_5				-1	1	
v_1v_3	1	1				
v_2v_4				-1	1	
v_1v_3	1	1				
v_2v_4			1	1		
v_0v_5					-1	-1

Figure 2.6

$$H_0(\triangle; \mathbb{F}) = \frac{ker(d_0)}{im(d_1)} \cong \frac{\mathbb{F}^6}{\mathbb{F}^5} \cong \mathbb{F}$$

$$H_1(\triangle; \mathbb{F}) = \frac{\ker(d_1)}{\operatorname{im}(d_2)} \cong \frac{\mathbb{F}^7}{\mathbb{F}^6} \cong \mathbb{F}$$

$$H_2(\triangle; \mathbb{F}) = \frac{\ker(d_2)}{\operatorname{im}(d_3)} \cong \frac{0}{0} \cong 0$$

$$H_n(\triangle; \mathbb{F}) = 0 \forall n \geq 3$$

We can use the fact that the rank of $H_0(\triangle; \mathbb{F})$, $H_1(\triangle; \mathbb{F})$ and $H_2(\triangle; \mathbb{F})$ equals the number of connected components, holes and voids of the simplicial complex $\triangle$ for simplicial complexes of maximal dimension 2 or for topological spaces that can be embedded in $\mathbb{R}^3$ (fulfilled in the last two examples) to cross check our results from computation.

Chapter 3

Persistence Theory and Persistence Diagrams

Persistence theory is the study of how long a feature of some topological data persists on the *timeline* of a *persistence parameter*. This persistence parameter may be time, elevation, resolution, etc. We will define persistence diagrams in the examples themselves. However, there are some basic terms that must be defined before moving on.

3.1 Basic Definitions

A persistence module is obtained from a nested family of topological spaces, whose intersection results in $\emptyset$ and union results in a maximal topological space (which we call X), *evolving* with the persistence parameter. This family is called a *filtration over X* . The homology functor is applied to this family pointwise and a persistence module is then formed with maps between two indices induced from the inclusion relation between them in the filtration, the collection of which is called *persistent homology*. More precise definitions for the two terms are given:

DEFINITION 3.1.1. A *filtration* χ over a subset of $\mathbb{R}$, T , is a family of topological spaces X_i , indexed by $i \in T$, such that $\forall i \leq j \in T, X_i \subseteq X_j$. If the filtration spaces X_i are simplicial complexes, then the filtration itself is called a *simplicial filtration*. We define a *finite simplicial filtration* in a similar manner (the X_i 's are finite simplicial complexes). A filtration over a topological space X has X_i s as subsets of X which satisfy the extra conditions $\bigcup_{i \in T} X_i = X$ and $\bigcap_{i \in T} X_i = \emptyset$ and the pair (X, χ) is called a *filtered space*.

DEFINITION 3.1.2. [2] Applying the homology functor to the sequence of spaces and maps in χ gives one persistence module per homology dimension p , denoted $H_p(\chi)$. The collection of these modules for p ranging over all dimensions is called the *persistent homology of χ* and denoted $H_*(\chi)$. Each module $H_p(\chi)$ has the homology group $H_p(X_i)$ as vector space at every index $i \in T$, and the homomorphism $H_p(X_i) \rightarrow H_p(X_j)$ induced by the inclusion map $X_i \hookrightarrow X_j$ at any pair of indices $i \leq j \in T$. The image $\text{im } H_p(X_i) \rightarrow H_p(X_j)$ is sometimes called a *p -th persistent homology group* of χ .

The last definition we make is of *sublevel filtration* and *superlevel filtration*.

DEFINITION 3.1.3. A *filter* of a topological space X is a function $f : X \rightarrow \mathbb{R}$. A filtration constructed over the family of sublevel sets $\mathcal{F}_i = f^{-1}(-\infty, i]$ is called a *sublevel filtration* while one constructed over the family of superlevel sets $\mathcal{F}^i = f^{-1}[i, \infty)$ is called a *superlevel filtration*.

Finally, we move on to examples.

3.2 Calculations

Example 3.2.1. We are given the dataset shown in Fig.3.1. The distance between adjacent points is 1 unit.

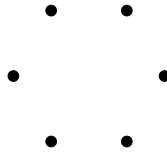


Figure 3.1: The dataset given

First, construct a filtration χ . For this, we introduce a notation $B_r(x)$, which refers to an open r -radius ball centered at the point x . Fix r . And now, construct a simplicial complex

$$\Delta(r) := \{Y \subseteq X \mid \bigcap_{x \in Y} B_r(x) \neq \emptyset\}$$

or in words, $Y \subseteq X$ is contained in $\Delta(r)$ if all the points in the convex hull of points in Y are contained in some $B_r(x), x \in Y$. This gives us a finite simplicial filtration (as there are only finite vertices) over $(S, \leq)$ since $r_1 \leq r_2 \implies \Delta(r_1) \subseteq \Delta(r_2)$. The geometric realization of these simplicial complexes forms a filtration over topological space x . We will only show the calculation for $H_0(\Delta(r); \mathbb{F})$, $H_1(\Delta(r); \mathbb{F})$ and $H_2(\Delta(r); \mathbb{F})$ for the case shown in Fig.3.2c. The rest cases and homology groups have similar calculations. Fix $r = R$ where $R \in (\sqrt{3}/2, 1]$.

$$\Delta(R) = \{\emptyset, v_0, \dots, v_5, v_0v_1, v_1v_2, \dots, v_0v_5, v_0v_2, \dots, v_0v_5, v_0v_1v_2, \dots, v_0v_1v_5\}$$

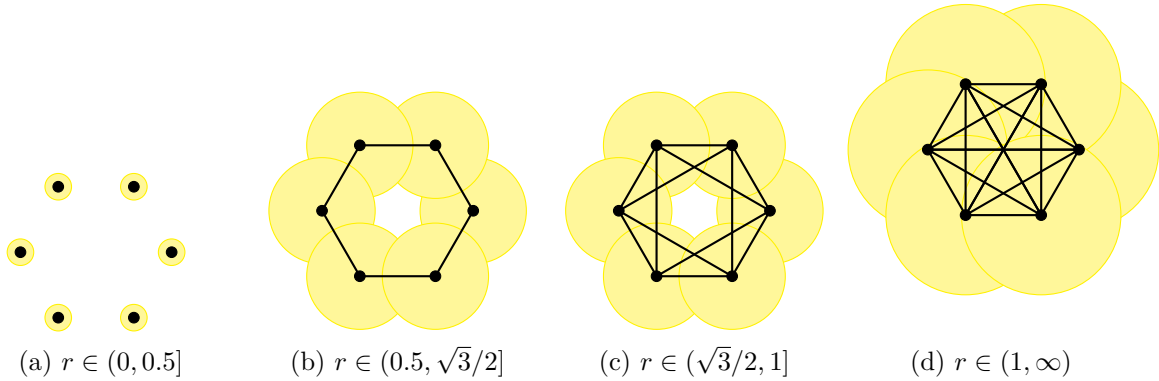


Figure 3.2

Next, write down the chain complex:

$$\mathbf{C}(\triangle(R)) : \cdots \xrightarrow{0} 0 \xrightarrow{0} \mathbb{F}^6 \xrightarrow{d_2} \mathbb{F}^{12} \xrightarrow{d_1} \mathbb{F}^6 \xrightarrow{0} 0 \xrightarrow{0} \cdots$$

From this, we conclude $\text{rank}(\ker(d_0)) = 6$, $\text{rank}(\text{im}(d_0)) = 0$ and $\text{rank}(\text{im}(d_3)) = 0$, $\text{rank}(\ker(d_3)) = 0$. Next, write down boundary matrix of d_1 .

$$\text{Boundary}(d_1) = \begin{bmatrix} -1 & & & & 1 & -1 & & \\ 1 & & & & & & & \\ 1 & -1 & & & & & -1 & \\ & 1 & & & & & & \\ & 1 & -1 & & & 1 & & -1 \\ & & 1 & -1 & & & 1 & -1 \\ & & & 1 & -1 & & & 1 \\ -1 & & & & & & & \\ & & & & 1 & -1 & & \\ & -1 & & & & & & 1 \end{bmatrix}$$

This can be reduced through column operations to:

$$\begin{bmatrix} -1 & -1 & & & \\ 1 & & -1 & & \\ & 1 & & -1 & \\ & & 1 & & -1 \\ & & & 1 & \\ & & & & 1 \end{bmatrix}$$

Hence, $\text{rank}(\text{im}(d_1)) = 5$ and $\text{rank}(\ker(d_1)) = 7$. Boundary matrix for d_2 is:

$$\text{Boundary}(d_2) = \begin{bmatrix} 1 & & & & & & 1 \\ 1 & 1 & & & & & \\ & 1 & 1 & & & & \\ & & 1 & 1 & & & \\ & & & 1 & 1 & & \\ & & & & 1 & 1 & 1 \\ -1 & & & & & 1 & 1 \\ & -1 & & & & & \\ & & -1 & & & & \\ & & & -1 & & & \\ & & & & -1 & & \\ & & & & & -1 & \\ & & & & & & -1 \end{bmatrix}$$

All the column vectors are linearly independent. Hence, $\text{rank}(\text{im}(d_2)) = 6$ and $\text{rank}(\ker(d_2)) = 6$. We get:

$$H_0(\triangle(R); \mathbb{F}) = \frac{\ker(d_0)}{\text{im}(d_1)} \cong \frac{\mathbb{F}^6}{\mathbb{F}^5} \cong \mathbb{F}$$

$$H_1(\Delta(R); \mathbb{F}) = \frac{\ker(d_1)}{\text{im}(d_2)} \cong \frac{\mathbb{F}^7}{\mathbb{F}^6} \cong \mathbb{F}$$

$$H_2(\Delta(R); \mathbb{F}) = \frac{\ker(d_2)}{\text{im}(d_3)} \cong \frac{\mathbb{F}^6}{\mathbb{F}^6} \cong \mathbb{F}$$

Calculating for the other cases, we get:

$$H_0(\chi) = \mathbb{I}(0, \infty) \oplus \mathbb{I}(0, 0.5] \oplus \mathbb{I}(0, 0.5] \oplus \mathbb{I}(0, 0.5] \oplus \mathbb{I}(0, 0.5] \oplus \mathbb{I}(0, 0.5]$$

$$H_1(\chi) = \mathbb{I}(0.5, 1]$$

$$H_2(\chi) = 0$$

The *decorated persistence diagram* is shown in Fig.3.3

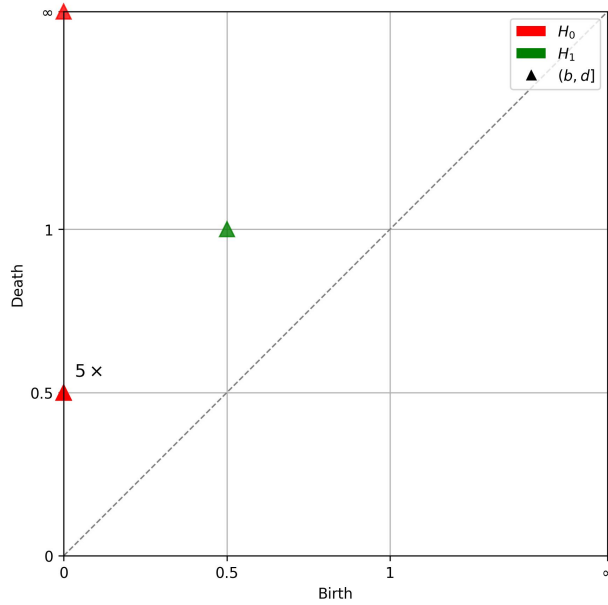


Figure 3.3: Decorated persistence diagram: A birth vs death plot of interval modules in the decomposition of persistence homology. Each color represents a persistence homology of degree n and each different shape represents a different kind of interval. Here, Δ stands for interval type $(b, d]$ where b =birth and d =death of interval. In undecorated persistence diagram, a point on the graph can represent any interval type.

Example 3.2.2. We are given the following topological space x with the axis beside denoting the height.

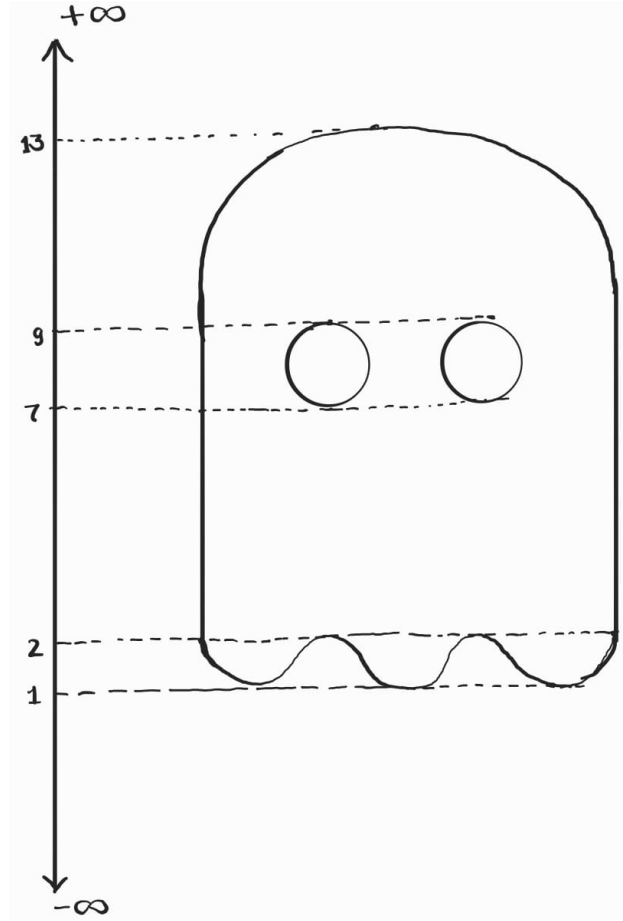


Figure 3.4: Topological space X (non-filled) with height axis

Take height function $f : X \rightarrow \mathbb{R}$ and construct a sublevel filtration χ . In this filtration, $\mathcal{F}_i$ denotes the portion of the topological space below height i . Since all we obtain are graph complexes, we can obtain rank of $H_0(\Delta; \mathbb{F})$ and $H_1(\Delta; \mathbb{F})$, where Δ is a triangulation of $\mathcal{F}_i$, by counting number of connected components and holes, respectively. $H_2(\Delta; \mathbb{F})$ will be trivial for the triangulation Δ of any $\mathcal{F}_i$. Note that we are only using this method as this has not been introduced before. We may use the method of constructing chain complex, boundary matrices and computing the rank of homology groups of degree n as well.

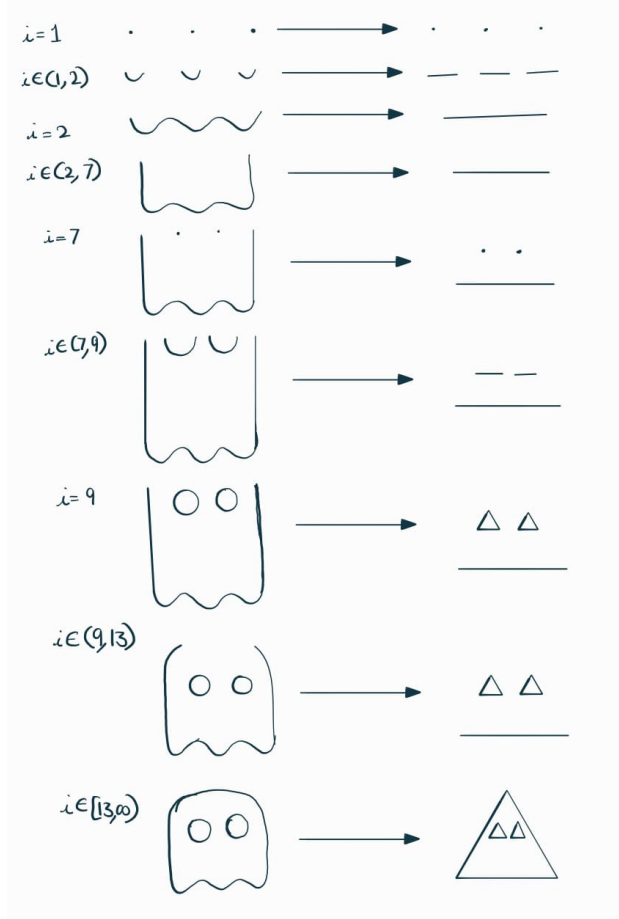


Figure 3.5: Evolution of X with height i (on right, the geometric realization of a triangulation of the portion)

Through Fig.3.6, we see that there are two connected components that live in $i \in [1, 2)$, one that lives in $i \in [1, \infty)$ and two more that live in $i \in [7, \infty)$. Similarly, there are two holes that live in $i \in [9, \infty)$ and one hole that lives in $i \in [13, \infty)$. Hence, we get:

$$H_0(\chi) = \mathbb{I}[1, \infty) \oplus \mathbb{I}[1, 2) \oplus \mathbb{I}[1, 2) \oplus \mathbb{I}[7, \infty) \oplus \mathbb{I}[7, \infty)$$

$$H_1(\chi) = \mathbb{I}[9, \infty) \oplus \mathbb{I}[9, \infty) \oplus \mathbb{I}[13, \infty)$$

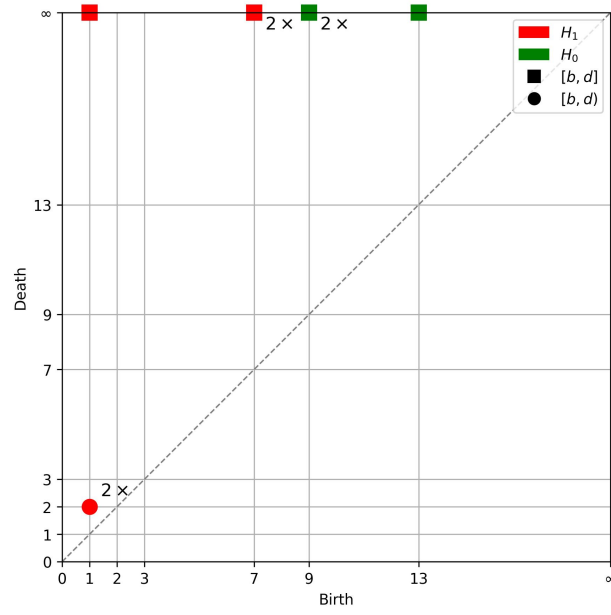


Figure 3.6: Decorated persistence diagram

References

- [1] James R. Munkres. *Elements of Algebraic Topology*. Addison-Wesley, 1984. ISBN: 0-201-04586-9.
- [2] Steve Y. Oudot. *Persistence Theory: From Quiver Representations to Data Analysis*. Vol. 209. Mathematical Surveys and Monographs. American Mathematical Society, 1979. ISBN: 978-1-4704-2545-6.